MS Applied Data Science

ADSP 31021 Machine Learning Operations
Assignment #1 - Data Versioning
Due Date – Beginning of Session #4

Assignment Overview

Modern machine learning systems require more than building predictive models, they require reproducibility, automation, version control, and operational thinking.
In this assignment, you will build a reproducible machine learning workflow that incorporates data versioning, automated pipelines, experimentation, and model evaluation using multiple dataset versions.
This assignment emphasizes production-oriented machine learning practices and introduces core MLOps principles used in real-world environments.

Instructions

1. Work individually unless otherwise approved.
2. Clearly label all sections and tasks in your notebook or scripts.
3. Include sufficient comments and markdown explanations throughout your work.
4. Submission: Please upload your completed solution on Canvas.
5. Your code must run end-to-end without manual intervention.
6. You may use Python, Jupyter notebooks, Scikit-learn, TensorFlow, PyTorch, XGBoost, Pandas, and other open-source libraries.
7. Total: 100 Points

Reproducibility Requirements

Your submission must include:

- requirements.txt or environment.yml
- Fixed random seed(s)
- Clearly documented execution steps
- Version-controlled datasets
- Reproducible train/test split logic

Required Tools

Required:

- DVC (Data Version Control)

Recommended:

- Git

• Jupyter Notebook or Python scripts
• Scikit-learn
• Pandas
• XGBoost
• TensorFlow / PyTorch (optional)
• Ruff / Flake8 / Pylint

Reproducibility Requirements

Your submission must include:

- requirements.txt OR environment.yml
- Fixed random seeds
- Version-controlled datasets
- Reproducible train/test split logic
- Clearly documented execution instructions
- Automated workflow execution

Dataset

Use the provided dataset: athletes.csv.zip

Dataset Version Definitions

You will create two dataset versions:

- **Dataset Version 1 (v1) -** The original/raw dataset after ingestion.
- **Dataset Version 2 (v2) -** A cleaned and transformed dataset after preprocessing.

Both versions must be tracked using the versioning tool.

Data Cleaning Requirements

Create dataset version v2 by applying cleaning and preprocessing operations such as:
- Removing invalid or missing values
- Removing outliers
- Cleaning survey response fields
- Creating new features
- Standardizing data types

Use the starter cleaning logic below as a baseline.

Clean data using following code:

Note: You may improve or extend this logic if properly documented and justified.

```
# Remove non-relevant columns
data = data.dropna(
    subset=[
        'region','age','weight','height','howlong',
        'gender','eat','background','experience',
        'schedule','deadlift','candj','snatch','backsq'
    ]
)

data = data.drop(
    columns=[
        'affiliate','team','name','athlete_id',
        'fran','helen','grace','filthy50',
        'fgonebad','run400','run5k','pullups','train'
    ]
)

# Remove outliers
data = data[data['weight'] < 1500]
data = data[data['gender'] != '--']
data = data[data['age'] >= 18]
data = data[(data['height'] < 96) & (data['height'] > 48)]

data = data[
    ((data['gender'] == 'Male') & (data['deadlift'] <= 1105)) |
    ((data['gender'] == 'Female') & (data['deadlift'] <= 636))
]

data = data[(data['candj'] > 0) & (data['candj'] <= 395)]
data = data[(data['snatch'] > 0) & (data['snatch'] <= 496)]
data = data[(data['backsq'] > 0) & (data['backsq'] <= 1069)]

# Clean survey data
decline_dict = {'Decline to answer|': np.nan}
data = data.replace(decline_dict)

data = data.dropna(
    subset=[
        'background','experience',
        'schedule','howlong','eat'
    ]
)
```

Required Workflow Tasks

Complete the following tasks for **both** selected data versioning tools.

Task 1 — Create Dataset Versions [10 Points]

Create:

- Dataset Version 1 (raw)

- Dataset Version 2 (processed)

Requirements:

- Track both datasets using DVC
- Demonstrate switching between versions
- Document versioning workflow

Deliverables:

- Dataset versions
- DVC commands used
- Evidence of successful version switching

Task 2 — Feature Engineering [10 Points]

Create the following feature:

```
df['total_lift'] = (
    df['deadlift'] +
    df['candj'] +
    df['snatch'] +
    df['backsq']
)
```

Deliverables:

- Documented preprocessing decisions
- Feature engineering explanation
- Final transformed dataset

Task 3 — Train/Test Split [10 Points]

Split both v1 and v2 into training and testing datasets using the same split ratio and fixed random seed.

Document:

- Split ratio
- Random seed
- Features used

Task 4 — Exploratory Data Analysis (EDA) [10 Points]

Perform EDA on:

- Dataset v1
- Dataset v2

Your EDA should include:

- Summary statistics
- Missing value analysis
- Distribution analysis
- Correlation analysis
- Visualizations
- Key observations

Deliverables:

- Visualizations
- Key findings
- Operational observations

Task 5 — Baseline Machine Learning Model [20 Points]
Using dataset v1:

1. Build a baseline regression model to predict total_lift
2. Clearly document:
    - Selected features
    - Model type
    - Training methodology
    - Assumptions

You may use:

- Linear Regression
- Random Forest
- XGBoost
- Neural Networks
- Other approved regression approaches

Task 6 — Model Evaluation [10 Points]
Evaluate the baseline model using at least:

- RMSE
- MAE
- R^2 Score

Document and interpret the results.

Task 7 - Build Reproducible Pipeline [10 Points]

Create a pipeline which includes the following stages

- Raw data Ingestion
- Data cleaning cleaning
- Feature engineering
- Model training
- Model evaluation

Task 8 — Retrain Using Dataset v2 [10 Points]
Without significantly changing the training pipeline:

1. Train the same model using dataset v2
2. Evaluate the updated model
3. Compare results against v1

Discuss:

- Accuracy differences
- Effects of cleaning
- Data quality impact
- Operational implications

Task 9 — Code Quality [10 Points]
Run a Python linter on your codebase.
Recommended tools:

- Pylint
- Flake8
- Ruff

Include:

- Linter output
- Summary of issues identified
- Improvements made

Deliverables

Code Deliverables

Submit:

- Jupyter notebook(s) or Python scripts

Your code must:

- Run successfully
- Be well organized
- Include comments/documentation

Dataset Deliverables
Submit:

- Dataset v1
- Dataset v2
- Evidence/screenshots of dataset versioning workflows

Discuss:

1. Which dataset version would you deploy and why?
2. How did dataset changes impact model performance?
3. What operational risks were discovered?
4. How does reproducibility improve ML systems?
5. What improvements would you make if deploying this system?